

JUNE 2026

Σ Telegraph

The Visa of Machine Intelligence:
A protocol for autonomous systems.

Validator Node Proposal



The Layer Where Intelligence Is Built, Bought, and Sold

Telegraph is the commercial rail for machine intelligence – the neutral layer where any service behind an API gets discovered, verified, and paid. Whether it's a model, dataset, tool, or live data feed, we don't own the supply. We are the payment and verification layer that autonomous agents and systems run on.

ANY SERVICE BEHIND AN API – PRODUCERS & DEMAND

Frontier
models

Data & API
providers

Academic labs

Indie
developers

Enterprises

Financial &
market data

Logistics &
compliance
tools

Autonomous
agents

Σ TELEGRAPH

The commercial & payment rail for machine intelligence

VERIFY · ROUTE · SETTLE · RECEIPT

END USERS & THE GLOBAL ECONOMY

The people and machines that consume intelligence

The verification, routing, and payment layer for machine intelligence

A permissionless protocol on Base that turns models, data, tools, and live feeds – from any provider – into verified, payable, auditable intelligence that autonomous agents can discover, pay for, and act on. The industry has the intelligence but no commercial layer for machines to trust and pay for it – no router, no trust, no standard, no recourse. Telegraph solves this.

How a request flows

- Plug in – any model, dataset, tool, or API joins as a miner with one config file.
- Request – an agent escrows USDC via x402.
- Route – sent to the best miners via a live leaderboard.
- Verify – Validators run evaluation scripts (written by Script Authors) to score Miner outputs against real ground truth during background checks and tournaments. Ground truth is proven with zkTLS.
- Settle – verified answer on-chain with a receipt; miner compensated in Machina via TWAP, validators earn emissions.

A neutral protocol that profits as the whole machine intelligence market grows

Four roles, kept strictly separate

Miners

Supply inference - compensated in Machina through the protocol's demand-driven TWAP mechanism (agents pay in USDC), with zero emissions.

Validators

Verify answers and reach consensus - earn emissions, plus a fee share.

Script authors

Write the open-source scoring scripts that validators run.

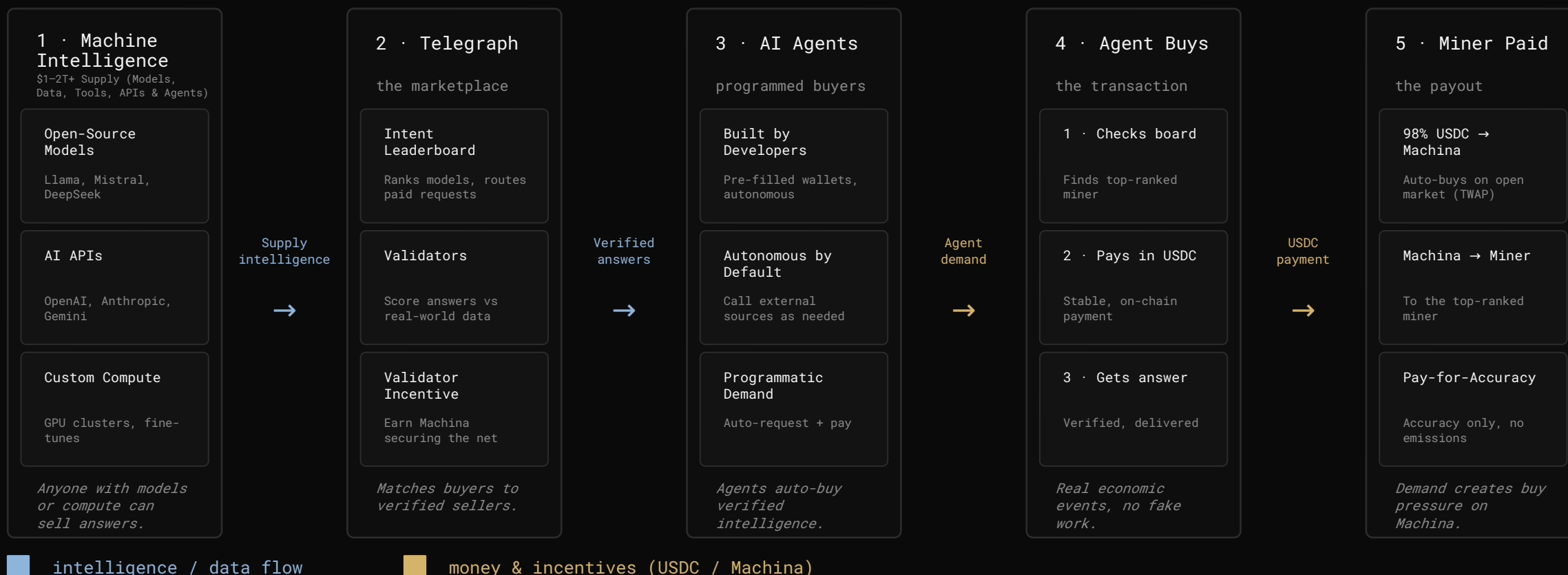
Agents

Consume verified answers and pay in USDC.

The edge: Telegraph builds, trains, and owns no models. Every new model funded anywhere in the world becomes a better supplier on the network at zero cost to the protocol.

The Telegraph ecosystem

How verified intelligence flows from suppliers (Miners) to autonomous agents



Why operating a node is a major opportunity

Operating a node makes you the gateway for your entire ecosystem - one of the fastest ways to build on and access the best verified intelligence on the market, while opening multiple new revenue streams from a single piece of infrastructure.

Lead your category

Be among the first to run verification infrastructure for the agent economy - a differentiated position as the fastest place to build on and reach the best intelligence available.

Arm your developers

One integration opens the entire intelligence supply to every builder in your network - verified answers from any model from a \$0.01 floor. The fastest way to ship agent-native apps.

Multiple revenue streams

Validator emissions plus a USDC fee share, a delegated-staking commission, and USDC from monetising your own models as miners - several income lines from one node.

Onboard your ecosystem

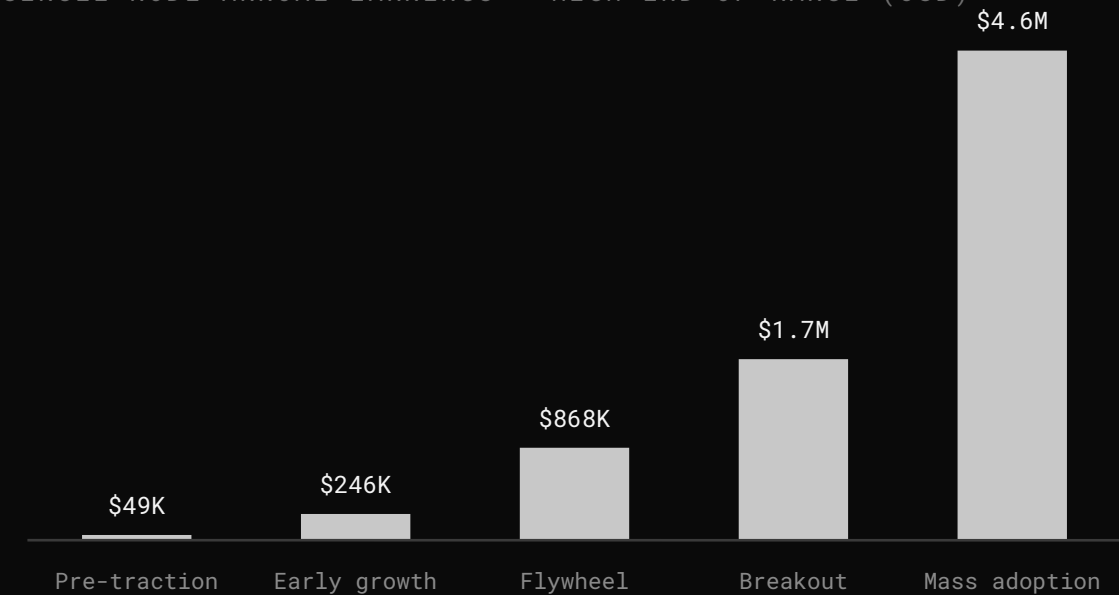
Bring developers, portfolio companies, and partners on as agents (demand) and miners (supply). Every connection you introduce routes through your node and compounds what it earns.



What a single node earns as the network grows

Earnings per node climb with request volume. The chart shows the high end of each stage; the table shows the full modelled range. Figures assume 64 nodes sharing the emission pool equally.

SINGLE-NODE ANNUAL EARNINGS – HIGH END OF RANGE (USD)



Network stage	Daily requests	Per node / yr
Pre-traction	~500	\$16K - \$49K
Early growth	~5,000	\$81K - \$246K
Flywheel active	~50,000	\$246K - \$868K
Breakout	~500,000	\$578K - \$1.66M
Mass adoption	2,000,000+	\$1.4M - \$4.6M

Illustrative projections derived from protocol parameters and the demand-multiplier schedule. Ranges reflect combined demand mechanics, validator staking, and market price. Not investment advice.

Run a node. Earn two ways.

The network is secured by 64 institutional validator nodes. As an operator, your node earns from two independent streams every day the network runs.

STREAM ONE

Protocol emissions

- 60% of 7,200 = 4,320 Machina/day, split equally across 64 nodes = 67.5 Machina per node per day.
- Approx. 24,637 Machina per node per year before the first halving - equal whether ranked 1st or 64th.
- Plus redistributed emissions, testing-cohort fee, script-author rewards, and delegated-staking commission.

STREAM TWO

USDC fee share (Roadmap)

- The protocol currently collects a 2% fee on every verified request, paid in USDC and directed to the Treasury.
- A future mechanism is planned to allocate up to 1% of this fee to active validators as USDC, providing a non-volatile revenue stream alongside Machina emissions.
- This allocation is not active at genesis and will be introduced strategically at a later stage based on network growth and validator participation.

Every active node earns the same emission share whether it ranks 1st or 64th - no need to become a whale to earn well.

Stop paying the intelligence tax.

Building AI in-house means training models, hiring scarce talent, renting GPUs, and managing vendors - before a single answer is even verified. Telegraph finds, routes, and sells verified intelligence from the entire market, so your node and your business pay only for the answer.

BUILD IT YOURSELF

\$78M – \$191M

to train one frontier model (GPT-4 to Gemini Ultra)

- AI / ML engineers at \$160K-\$300K each, per year
- \$500K+ for a single 8-GPU server, plus power and ops
- 80%+ of AI projects fail to reach value (RAND)
- You still build verification and monitoring yourself

BUILD ON TELEGRAPH

\$0.01

per verified request - from the entire market

- No training, no GPUs, no ML team to hire
- Auto-routed to the best model on Earth
- Verification and a cryptographic receipt included
- Scale instantly; pay only for the answer

Intelligence keeps getting cheaper. Inference costs have fallen ~280x in 18 months - from \$20 to \$0.07 per million tokens. Telegraph aggregates that competition and sells you the winning answer.

Create and sell your own intelligence through Σ Telegraph

A node is not just a buyer of intelligence - it is a seller. Register your own models or subnets as miners and turn them into a revenue line, earning USDC from global agent demand on top of whatever you already make from them internally.

1 Register as a miner

Plug any model, dataset, tool, or API in with one config file. No retraining, no migration - it competes on the live leaderboard from day one.

2 Compete on accuracy

Routing is pure pay-for-accuracy. The best answer wins the request - your model earns purely on merit, scored against real ground truth.

3 Earn USDC on demand

Every winning request settles in USDC, converted to Machina via TWAP. Global agent demand becomes a direct, on-chain revenue stream.

4 Earn twice

Monetise the same models across both your internal products and the open Telegraph market - one asset, two income lines.

Unlock new revenue across your network

Operating a node makes you the gateway for everyone you work with - your developers, portfolio companies, partners, and users - to plug in. Every connection you bring compounds what your node earns.

Validator yield

Two streams - emissions plus a fee share - on every request the network settles. The base return for operating a node.

Delegated staking

Holders across your network delegate their Machina to your node. You take a declared commission on their share of emissions.

Monetize your own models

Register your models or subnets as miners and earn USDC from agent demand, on top of revenue you already make. Earn twice.

Onboard your ecosystem

Bring your developers and businesses on as agents and miners. Activity you introduce flows through the node you operate.

Keys, ground truth, consensus, and slashing

Key separation

Manager Key (cold) holds stake offline; Operator Key (hot) only signs. A breach can't touch principal; keys rotate on-chain.

Ground truth (zkTLS)

The leader proves scraped data with zero-knowledge TLS; all validators verify for free; fabricated proofs are rejected.

Consensus (BFT)

43 of 64 must sign to finalise (tolerates 21 bad/offline); stake-weighted median, commit-reveal, BLS-aggregated.

Credentials

Miner API keys encrypted across hardware enclaves; no single node holds the full key; auto-rotated on secure hardware.

Slashing

Liveness failure = ejection (no burn); proven fraud = 20% slash + ban; deviation forfeits emissions only, principal untouched.

Abuse resistance

Economic Sybil cost on a fixed 21M supply; jittered TWAP settlement makes front-running unprofitable.

Fixed, Bitcoin-style, demand-linked

- 21,000,000 Machina hard cap; 7,200 per epoch genesis emission; halving every 4 years; zero pre-mine, zero team / VC allocation.
- Emission split (immutable): Validators 60% / Script authors 20% / Treasury 20%; miners receive 0% emissions (USDC converted to Machina).
- Value linkage - every paid request converts USDC into Machina via continuous TWAP, so real usage is a mechanical market buy.
- Immutable forever: cap, halving, 60/20/20 split, zero pre-mine; other parameters need 43-of-64 plus a timelock.

21M

hard cap, fixed

60 / 20 / 20

validator / script / treasury

0%

miner emissions

Risk profile

- Token price - emissions and staked principal are valued in a new token with thin early liquidity.
- Staked capital - 21-day unbonding; 20% slash on proven fraud; the slot is held by staying top-64, so stake may need topping up. Active node partners can mitigate this through a set allocation of Machina from protocol treasury emissions.
- Stage and team - early protocol, small core team.

Telegraph works with serious operators to structure entry - including treasury support for active node partners - so the capital and slot mechanics are manageable rather than a barrier.

What it costs and how to deploy

PRE-MAINNET (CURRENT PHASE)

Running a node during testnet is extremely lightweight.

- Infrastructure - a standard server with 8-16 GB RAM; no GPUs or specialized hardware required.
- Cost - low hundreds of USD per month, plus a couple of testnet tokens.
- Setup - run a single script and bind a private key; no staking or capital at this stage.
- Staff - minimal oversight (uptime and key management).

Run a node immediately with almost zero financial commitment or operational complexity.

POST-MAINNET

Requirements shift to operating as a full validator.

- Capital - stake enough Machina to remain in the top 64 (market-determined, 21-day unbonding).
- Infrastructure - still lightweight, no GPUs; a secure enclave-capable server; submission gas reimbursed.
- Setup - stake via Manager Key (cold), bind an Operator Key (hot), deploy one config file, sync to NTP.
- Staff - minimal; uptime monitoring and basic key hygiene.

Compound further through delegated staking, running your own miners, and onboarding ecosystem agents.



Σ Telegraph

Email: info@telegraphprotocol.com

Web: <https://telegraphprotocol.com>